

20 Detailed Easy–Moderate Software Hackathon Projects

Cybersecurity • FinTech • EdTech • AI • GenAI • Automation • GIS

Project-planning handbook for a student hackathon team

Each idea includes the problem, use case, need, source/challenge direction, unique approach, MVP scope, advanced extensions, suggested stack, demo plan, risks and an unofficial potential score.

Important source note

The project titles in this document are **proposed/adapted ideas**. They are not presented as official SIH 2026 or Hack Celestia problem-statement titles. “Source” identifies the challenge/theme that inspired the direction. Official SIH problem statements should be verified against the current catalogue before submission.

Prepared as a practical idea-selection and prototype-planning document.

1. Source & Challenge Context

These projects are mapped primarily to challenge directions published on Hack Celestia 2026. The site lists domains including Agentic AI, Cyber Security, FinTech & Digital Economy, Generative AI, Human–AI Collaboration, IoT/Edge AI, Pharma/Biotech, Metaverse/VR and SpaceTech/Satellite Applications. The source mapping in this PDF uses those published themes as inspiration rather than claiming that the proposed project names are official problem statements. ■cite■turn1search1■

SIH 2026 is also a useful benchmark for software-hackathon scope. BlinkNBuild's current SIH 2026 portal describes 236 problem statements, including 178 software and 58 hardware entries, with a ■1,00,000 prize per winning problem statement. Its page also notes that the live interactive repository displays 226 entries, so the catalogue count should be checked against the latest official release when selecting an exact SIH problem statement.

■cite■turn1search0■

How to read the “Potential” number

The 83–93/100 figures are **unofficial project-potential estimates** created for this handbook. They are not official judging scores, not predictions of winning, and not scores assigned by Hack Celestia or SIH. They are intended only to help compare prototype scope, demo clarity, technical depth, measurable evaluation and practical impact.

Recommended selection rule

Pick the idea for which your team can build a convincing end-to-end demo, collect at least one small evaluation dataset, explain the architecture, and show a clear human/user benefit. A narrower working prototype is generally more useful for a hackathon than a huge feature list that cannot be demonstrated.

2. Index

ID	Project	Domain	Difficulty	Potential
P01	PhishGuard — Explainable Phishing Detection	Cybersecurity + AI	5/10	90/100
P02	SafeAI — AI Security Test Lab	Cybersecurity + AI	6/10	93/100
P03	LogLens — AI Security Log Analyzer	Cybersecurity + Automation	6/10	89/100
P04	ScamShield — Scam Message Analyzer	Cybersecurity + GenAI	5/10	88/100
P05	BudgetPilot — AI Spending Coach	FinTech + AI	5/10	88/100
P06	FraudFlag — Transaction Anomaly Detector	FinTech + AI	6/10	91/100
P07	FinDoc — Smart Invoice Extractor	FinTech + Automation	5/10	86/100
P08	StudyFlow — Adaptive AI Study Planner	EdTech + GenAI	5/10	87/100
P09	CampusRAG — College Knowledge Assistant	EdTech + GenAI	5/10	89/100
P10	SkillGap AI — Resume-to-Job Analyzer	EdTech + AI	5/10	85/100
P11	ExamSense — Weak Topic Detector	EdTech + AI	6/10	88/100
P12	AutoRoute — Complaint & Ticket Automation	Automation + AI	5/10	86/100
P13	Meeting2Action — Meeting-to-Task Automation	Automation + GenAI	5/10	84/100
P14	DocuFlow — Smart Form Filling Assistant	Automation + AI	6/10	87/100
P15	SafeRoute — Multi-Factor Route Planner	AI + GIS + Automation	6/10	88/100
P16	OpsPilot — AI Workflow Automation Builder	Automation + Agentic AI	6/10	92/100
P17	AlertHub — Smart Notification Prioritizer	AI + Automation	5/10	83/100
P18	AccessGuard — Risk-Based Login Simulator	Cybersecurity + FinTech	6/10	90/100
P19	ScholarshipMatch — AI Eligibility Assistant	EdTech + FinTech + AI	5/10	86/100
P20	CrisisAssist — Emergency Response Dashboard	Agentic AI + Automation	6/10	92/100

The detailed sections begin on the next page. Each project is designed to be buildable at an easy–moderate level while leaving room for a technically stronger extension.

P01 — PhishGuard — Explainable Phishing Detection

Domain	Cybersecurity + AI	Difficulty	5/10
Estimated potential	90/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Cyber Security: Autonomous Threat Detection & Response; Building Trust in the Digital Future.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

Students, employees and everyday users receive suspicious emails, SMS messages and URLs but often cannot tell exactly what makes a message dangerous. A binary “safe/unsafe” label is not enough because users need understandable evidence before deciding what to do.

Use case / proposed solution

A web application where a user pastes a message, email body or URL. The system extracts indicators such as urgency, impersonation, credential requests, suspicious domains, mismatched links and unusual language. It produces a risk level, evidence cards and safe next actions.

Why do we need it?

Phishing succeeds partly because users make decisions under time pressure. An explainable interface can turn a technical detection result into a practical decision aid and can also be used for awareness training.

Unique approach

Use a layered detector: deterministic URL/domain rules first, lightweight NLP features second, and an optional LLM explanation layer last. Keep evidence separate from generated explanation so the UI can show exactly which signals triggered the warning.

Prototype workflow

- Input email/SMS/URL
- Normalize and extract links
- Run URL and text risk rules
- Generate feature-level evidence
- Calculate risk score
- Show explanation and recommended safe action
- Allow user feedback for later evaluation

Minimum viable prototype (MVP)

- Paste text/URL
- Risk score with Low/Medium/High bands
- 5–10 explainable indicators
- Safe-action checklist
- History of analyzed samples
- Demo dataset containing benign and phishing examples

Advanced extensions

- QR/link screenshot analysis
- Brand impersonation detection
- Browser extension
- Organization dashboard
- Feedback-based threshold tuning
- Multilingual scam/phishing analysis

Suggested stack

React or Next.js; FastAPI; Python; URL parsing; regex/rules; optional transformer/LLM; SQLite/PostgreSQL.

Live demo plan

Show three messages: a normal college notice, a fake bank alert and a fake scholarship message. Compare the evidence cards and demonstrate that the system explains **why** the risk changed.

Risks / limitations

False positives can reduce trust. Do not claim perfect detection. Use synthetic/public examples for the prototype and clearly label the system as decision support.

How to make the project measurable

Strong demo value comes from explainability, a visible security workflow, measurable precision/recall on a small benchmark, and a clear user action after detection.

P02 — SafeAI — AI Security Test Lab

Domain	Cybersecurity + AI	Difficulty	6/10
Estimated potential	93/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Cyber Security: Securing the AI-Driven Digital Ecosystem.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

AI applications can fail in ways traditional software tests do not capture. Prompt injection, instruction leakage, unsafe requests and conflicting instructions can expose weaknesses before a model-powered application reaches users.

Use case / proposed solution

A controlled testing dashboard that sends predefined adversarial test cases to a model endpoint, records responses, classifies observed weaknesses and generates a security report.

Why do we need it?

Teams often demonstrate that an AI system works but do not demonstrate how it behaves under hostile input. A repeatable security test harness makes AI safety measurable and easier to regression-test.

Unique approach

Create a library of test categories, run them automatically, compare responses with expected safety criteria, and present a per-test severity. Separate raw model output from the evaluator's explanation.

Prototype workflow

- Choose model/app endpoint
- Select test suite
- Execute test cases
- Capture prompt/response
- Evaluate against rules
- Assign severity
- Generate mitigation report
- Re-run after changes

Minimum viable prototype (MVP)

- 20–30 controlled test cases
- Prompt-injection category
- Instruction-leakage category
- Unsafe-output category
- Pass/fail dashboard
- Exportable report

Advanced extensions

- Model-to-model comparison

- Custom policy editor
- Regression testing
- Red-team/blue-team mode
- Risk trend over versions
- Local-model support

Suggested stack

React; FastAPI; Python; model API/local model; JSON test suites; SQLite; optional LLM-as-judge with deterministic checks.

Live demo plan

Run the same test suite against a deliberately weak demo chatbot and a hardened version. Show which cases changed from fail to pass after a mitigation.

Risks / limitations

Avoid real-world abuse targets. Keep tests inside a sandbox and use harmless payloads designed only to evaluate the demo application.

How to make the project measurable

The idea can demonstrate novelty through a concrete testing workflow, repeatability, before/after evidence, and a clear security report rather than a generic chatbot.

P03 — LogLens — AI Security Log Analyzer

Domain	Cybersecurity + Automation	Difficulty	6/10
Estimated potential	89/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Cyber Security: Autonomous Threat Detection & Response.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Security logs are noisy and difficult to inspect manually. Important patterns may be distributed across many events, such as repeated login failures followed by a successful login or unusual access from a new source.

Use case / proposed solution

Upload CSV/JSON logs and receive grouped incidents, anomaly indicators and an evidence timeline. The tool should explain which events contributed to each alert.

Why do we need it?

A small organization may not have a full SOC. A lightweight analyzer can demonstrate the core idea of detection, triage and explanation without requiring enterprise infrastructure.

Unique approach

Normalize heterogeneous log formats, create event fingerprints, calculate rule-based anomaly signals and optionally use clustering/ML for unusual sequences. Present an incident timeline rather than isolated alerts.

Prototype workflow

- Upload logs
- Normalize fields
- Create event fingerprints
- Detect rules/anomalies
- Group related events
- Build incident timeline
- Summarize evidence
- Mark analyst disposition

Minimum viable prototype (MVP)

- CSV/JSON upload
- Login-failure detection
- New-IP detection
- Burst detection
- Incident timeline
- Search/filter
- Severity labels

Advanced extensions

- Sequence anomaly model
- MITRE-style technique mapping
- Streaming mode
- Alert deduplication
- Analyst feedback loop
- Role-based dashboard

Suggested stack

Python; Pandas; FastAPI; React; SQLite/PostgreSQL; scikit-learn optional.

Live demo plan

Load a synthetic 5,000-event log file containing one suspicious sequence. Show how hundreds of events collapse into one understandable incident.

Risks / limitations

Synthetic logs are safer and easier to reproduce. Do not imply the prototype is a production SIEM.

How to make the project measurable

Evaluation can be strengthened with a labeled synthetic dataset and metrics such as detection precision, recall and time-to-triage.

P04 — ScamShield — Scam Message Analyzer

Domain	Cybersecurity + GenAI	Difficulty	5/10
Estimated potential	88/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Cyber Security: Building Trust in the Digital Future; Generative AI: Responsible & Trustworthy Generative AI.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

Scam messages often rely on behavioral pressure: urgency, fear, rewards, authority impersonation and requests for OTPs, payments or personal data. Users may recognize the message as suspicious but still struggle to identify the tactic.

Use case / proposed solution

Paste a message or call transcript and receive a scam-risk assessment, detected persuasion tactics, requested information, suspicious links and a safe-response checklist.

Why do we need it?

The useful output is not only detection. The system teaches the user what pattern to look for next time, making it suitable for digital-safety awareness.

Unique approach

Build a taxonomy of scam tactics and combine deterministic patterns with an NLP/LLM classifier. For each detected tactic, display the exact supporting phrase rather than only a generic explanation.

Prototype workflow

- Paste message/transcript
- Extract entities and requests
- Match scam-tactic rules
- Run optional classifier
- Generate evidence cards
- Recommend safe actions
- Store anonymous feedback

Minimum viable prototype (MVP)

- Text input
- Risk band
- Tactic detection
- Evidence highlighting
- Safe-action checklist
- Example library

Advanced extensions

- Regional language support

- Voice transcription
- Screenshot OCR
- Family/elder-friendly mode
- Awareness quiz mode

Suggested stack

React; FastAPI; Python; OCR/speech-to-text optional; rules + NLP/LLM.

Live demo plan

Use a fake delivery-payment message, a fake internship offer and a normal message. Show different tactics and explain why the scores differ.

Risks / limitations

Do not collect real banking credentials, OTPs or personal secrets. Use fabricated examples.

How to make the project measurable

A strong prototype combines social-impact clarity, explainability and a practical user flow that ends in a safe action.

P05 — BudgetPilot — AI Spending Coach

Domain	FinTech + AI	Difficulty	5/10
Estimated potential	88/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — FinTech & Digital Economy: Intelligent Financial Decision Support.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Transaction statements contain useful information but raw rows do not immediately answer questions such as “Where did my spending increase?” or “Which recurring expenses are growing?”

Use case / proposed solution

Upload a synthetic or redacted CSV statement. The application categorizes transactions, identifies recurring expenses, compares periods and explains the main drivers of change.

Why do we need it?

Financial dashboards often show totals without explaining the reasons behind changes. A decision-support tool should connect a summary to the underlying transactions.

Unique approach

Use a transparent categorization pipeline with merchant rules, category mapping and optional NLP. Every insight should link back to the transactions that support it.

Prototype workflow

- Upload CSV
- Normalize transactions
- Categorize merchants
- Detect recurring items
- Compare periods
- Generate insights
- Show supporting transactions

Minimum viable prototype (MVP)

- CSV upload
- Category breakdown
- Monthly comparison
- Recurring expense detection
- Top spending drivers
- Simple budget view

Advanced extensions

- What-if simulator
- Goal planning
- Anomaly alerts

- Natural-language questions
- Privacy-first local processing

Suggested stack

React; FastAPI; Pandas; SQLite; optional LLM for explanation.

Live demo plan

Upload two months of synthetic data and ask why spending increased. Show the top three transaction groups responsible.

Risks / limitations

Use synthetic or anonymized data. Avoid investment, lending or tax advice claims.

How to make the project measurable

The strongest differentiator is evidence-linked explanations instead of a generic personal-finance chatbot.

P06 — FraudFlag — Transaction Anomaly Detector

Domain	FinTech + AI	Difficulty	6/10
Estimated potential	91/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — FinTech & Digital Economy: AI-Driven Financial Risk & Fraud Intelligence.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Fraud patterns are often context-dependent. A large transaction is not automatically suspicious, while a smaller transaction may be unusual when it occurs at a new merchant, unusual time or after a rapid sequence of transactions.

Use case / proposed solution

Generate or upload synthetic transactions and score them for review. The system explains which features caused the anomaly score and places suspicious cases in a review queue.

Why do we need it?

A review-oriented prototype is more realistic than claiming fully automatic fraud blocking. Human analysts can inspect evidence before a decision.

Unique approach

Combine statistical baselines with an anomaly model. Normalize transaction behavior per synthetic customer profile and expose feature contributions in plain language.

Prototype workflow

- Create/import transactions
- Build customer baselines
- Calculate anomaly features
- Score transactions
- Create review queue
- Show explanation
- Mark analyst decision

Minimum viable prototype (MVP)

- Synthetic data generator
- Anomaly model
- Review queue
- Feature explanation
- Basic metrics

Advanced extensions

- Graph-based merchant relationships
- Concept drift simulation
- Threshold tuning
- Case management

- Model comparison

Suggested stack

Python; Pandas; scikit-learn; FastAPI; React; SQLite/PostgreSQL.

Live demo plan

Generate a normal customer history and inject a few controlled anomalies. Show how the model detects them and which features contributed.

Risks / limitations

Fraud detection requires careful thresholding and fairness analysis. Keep the prototype synthetic and avoid real payment data.

How to make the project measurable

A reproducible synthetic benchmark and explainable review queue make the demo measurable and credible.

P07 — FinDoc — Smart Invoice Extractor

Domain	FinTech + Automation	Difficulty	5/10
Estimated potential	86/100	Source type	Adapted project idea; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — FinTech & Digital Economy: Intelligent Financial Decision Support; adapted toward document automation.

Adapted project idea; not an official challenge title.

What is the problem?

Invoices contain semi-structured fields that finance teams repeatedly copy into systems. Manual entry is slow and can introduce errors.

Use case / proposed solution

Upload an invoice image or PDF and extract vendor, invoice number, date, tax, subtotal and total. Highlight uncertain fields and require confirmation before saving.

Why do we need it? Automation is useful only when users can see and correct mistakes. Confidence-aware extraction is safer than silently filling fields.	Unique approach Use OCR/PDF text extraction followed by field-specific parsing. Use validation rules such as $\text{total} = \text{subtotal} + \text{tax}$ where applicable and highlight conflicts.
--	--

Prototype workflow

- Upload document
- Extract text
- Detect fields
- Validate values
- Highlight confidence
- User confirms
- Save structured record

Minimum viable prototype (MVP)

- PDF/image upload
- Core field extraction
- Confidence indicator
- Edit/confirm screen
- CSV/JSON export

Advanced extensions

- Batch processing
- Duplicate invoice detection
- PO matching

- Expense categorization
- Accounting-system connector

Suggested stack

Python; OCR; FastAPI; React; SQLite; optional document LLM.

Live demo plan

Use three invoice templates with different layouts and show that the extraction interface still produces a common structured format.

Risks / limitations

OCR errors are common. Always provide a correction screen and avoid claiming accounting compliance.

How to make the project measurable

A polished document workflow with validation is more compelling than a raw OCR demo.

P08 — StudyFlow — Adaptive AI Study Planner

Domain	EdTech + GenAI	Difficulty	5/10
Estimated potential	87/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to education.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Static study timetables fail when students miss sessions, discover weak topics or have changing availability. The plan should adapt rather than simply display a calendar.

Use case / proposed solution

Student enters syllabus/topics, exam date, available hours and confidence levels. The planner creates a schedule, tracks completion and reallocates missed work without creating an impossible backlog.

Why do we need it?

Adaptation is the central value. The system should respond to evidence from quizzes and completed sessions rather than only generating a one-time timetable.

Unique approach

Use a deterministic scheduling engine for workload allocation and an optional LLM for explanations. Keep scheduling constraints explicit and testable.

Prototype workflow

- Enter syllabus
- Set exam date/hours
- Estimate topic difficulty
- Generate plan
- Track sessions
- Collect quiz feedback
- Rebalance future sessions

Minimum viable prototype (MVP)

- Topic input
- Calendar plan
- Completion tracking
- Missed-session recovery
- Progress dashboard

Advanced extensions

- Automatic syllabus parsing
- Quiz integration
- Energy/time preference modeling
- Spaced repetition

- Teacher dashboard

Suggested stack

React; FastAPI; Python scheduling rules; optional LLM; SQLite.

Live demo plan

Create a 14-day plan, mark two sessions missed, then show the revised plan and explain the changes.

Risks / limitations

Do not overload students with unrealistic schedules. Allow manual override and show assumptions.

How to make the project measurable

A visible before/after adaptation is an excellent live demo because judges can interact with the system.

P09 — CampusRAG — College Knowledge Assistant

Domain	EdTech + GenAI	Difficulty	5/10
Estimated potential	89/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Generative AI: AI-Powered Digital Knowledge Systems; adapted to education.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

College information is scattered across notices, handbooks, PDFs and web pages. Students repeatedly ask the same questions and generic chatbots can hallucinate answers.

Use case / proposed solution

Upload college documents and ask questions. The assistant retrieves relevant passages, answers from them and shows the source document/page or excerpt used.

Why do we need it?

Grounding and citation are important in institutional assistants. If evidence is missing, the system should say so instead of inventing an answer.

Unique approach

Use retrieval-augmented generation with chunking, embeddings and metadata filters. Add a confidence/evidence gate before answer generation.

Prototype workflow

- Upload approved documents
- Extract and chunk text
- Create embeddings
- Retrieve top passages
- Generate grounded answer
- Show citations
- Log unanswered questions

Minimum viable prototype (MVP)

- PDF upload
- Document indexing
- Question answering
- Source excerpts
- No-evidence response

Advanced extensions

- Admin document approval
- Multilingual queries
- Notice change detection
- Role-based access

- Analytics dashboard

Suggested stack

React; FastAPI; embeddings; vector DB; LLM; PDF parser; SQLite/PostgreSQL.

Live demo plan

Upload a college handbook and several notices, then ask questions whose answers exist and one whose answer does not.

Risks / limitations

Only use authorized documents. Access controls are important if documents contain private information.

How to make the project measurable

The prototype should emphasize grounded answers, source visibility and safe handling of unknown questions.

P10 — SkillGap AI — Resume-to-Job Analyzer

Domain	EdTech + AI	Difficulty	5/10
Estimated potential	85/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to careers.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Students often receive a job description and a resume but cannot easily identify which requirements are covered, missing or weakly evidenced.

Use case / proposed solution

Upload a resume and job description. Produce a requirement-to-evidence map, skill gaps and a small set of practical projects or learning actions to close the gaps.

Why do we need it? The useful output is a structured plan, not just a similarity score. Students can see exactly which requirement is unsupported by their resume.	Unique approach Extract skills and requirements, normalize synonyms, map evidence sentences to requirements and generate a gap report. Keep the evidence mapping visible.
--	---

Prototype workflow

- Parse resume
- Parse job description
- Normalize skills
- Map evidence
- Identify gaps
- Suggest projects/actions
- Export report

Minimum viable prototype (MVP)

- PDF/text resume input
- Job-description input
- Skill map
- Gap report
- Project suggestions

Advanced extensions

- Skill graph
- Portfolio/GitHub evidence
- Role-specific roadmaps
- Multiple resume versions

- ATS-style formatting checks

Suggested stack

React; FastAPI; PDF parser; embeddings/NLP; optional LLM.

Live demo plan

Use a sample student resume and a software internship description. Show covered requirements, partial matches and missing skills.

Risks / limitations

Do not present the tool as an employer selection system. Avoid sensitive personal attributes in scoring.

How to make the project measurable

A transparent evidence map is more defensible than a single opaque compatibility percentage.

P11 — ExamSense — Weak Topic Detector

Domain	EdTech + AI	Difficulty	6/10
Estimated potential	88/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to learning analytics.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

A test score alone does not tell a student why they are losing marks. Errors can come from concept gaps, question-type weaknesses or repeated careless mistakes.

Use case / proposed solution

Import topic-tagged quiz results and classify weak areas. The dashboard separates concept-level weaknesses from error patterns and recommends targeted practice.

Why do we need it?

Students need actionable feedback. The project demonstrates how analytics can turn raw quiz events into a personalized intervention.

Unique approach

Track topic accuracy, difficulty, time and error type. Use a simple model/rules first; add ML only when the dataset supports it.

Prototype workflow

- Import quiz results
- Tag questions/topics
- Compute performance features
- Identify weak patterns
- Recommend practice
- Track improvement

Minimum viable prototype (MVP)

- CSV quiz import
- Topic dashboard
- Weak-topic ranking
- Error-type analysis
- Practice recommendations

Advanced extensions

- Adaptive question generation
- Knowledge graph
- Difficulty calibration
- Teacher dashboard

- Longitudinal progress

Suggested stack

React; Python; Pandas; FastAPI; SQLite; optional ML/LLM.

Live demo plan

Inject a dataset where a student has good overall accuracy but poor performance on two specific concepts. Show targeted recommendations.

Risks / limitations

Small datasets can make ML unstable. Make the rules transparent and show sample size.

How to make the project measurable

The project becomes stronger when recommendations are tied directly to measurable mistakes and later improvement.

P12 — AutoRoute — Complaint & Ticket Automation

Domain	Automation + AI	Difficulty	5/10
Estimated potential	86/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Agentic AI: Autonomous Enterprise Operations.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Organizations receive many support tickets that must be categorized, prioritized and routed. Manual triage delays response and can produce inconsistent routing.

Use case / proposed solution

Accept a ticket and automatically infer category, location/team, urgency and SLA. Show the evidence behind the routing decision and escalate overdue items.

Why do we need it? This is a realistic automation problem with a clear human-in-the-loop boundary. The system can save time without pretending to replace staff.	Unique approach Use structured categories, classifier/rules and an optional LLM for unstructured text. Keep routing policies editable by administrators.
--	--

Prototype workflow

- Receive ticket
- Extract fields
- Classify category
- Estimate urgency
- Select team
- Set SLA
- Track status
- Escalate if needed

Minimum viable prototype (MVP)

- Ticket form
- Automatic category
- Priority
- Routing queue
- SLA timer
- Audit trail

Advanced extensions

- Email ingestion
- Duplicate detection

- Agentic multi-step routing
- Analytics
- Policy editor

Suggested stack

React; FastAPI; classifier/LLM; PostgreSQL/SQLite.

Live demo plan

Submit five different complaints and show automatic routing, then edit a routing policy and rerun the same ticket.

Risks / limitations

Wrong routing can delay service. Include human override and audit logs.

How to make the project measurable

Policy transparency and an end-to-end workflow make this stronger than a simple text classifier.

P13 — Meeting2Action — Meeting-to-Task Automation

Domain	Automation + GenAI	Difficulty	5/10
Estimated potential	84/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Agentic AI: Autonomous Enterprise Operations; Human–AI Collaboration: AI as a Collaborative Intelligence Partner.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

Meeting notes often contain decisions, tasks and deadlines mixed with general discussion. Important actions can be forgotten after the meeting.

Use case / proposed solution

Process a transcript and extract decisions, action items, owners and deadlines. Present a review screen before creating tasks.

Why do we need it? The human approval step addresses a major weakness of autonomous meeting agents: incorrect attribution or invented commitments.	Unique approach Classify sentences into discussion, decision, action and context. Extract task fields and attach evidence snippets from the transcript.
--	---

Prototype workflow

- Upload transcript/audio
- Transcribe if needed
- Segment speaker turns
- Extract decisions/tasks
- Attach evidence
- Human review
- Export tasks

Minimum viable prototype (MVP)

- Transcript upload
- Action extraction
- Owner/deadline extraction
- Evidence snippet
- Approve/edit screen

Advanced extensions

- Calendar integration
- Task-manager connector
- Meeting trend analytics

- Multilingual support

Suggested stack

Speech-to-text; FastAPI; LLM; React; SQLite.

Live demo plan

Use a short fictional project meeting with three action items and show how the system extracts them with evidence.

Risks / limitations

Speaker attribution can be imperfect. Never silently assign high-impact actions without confirmation.

How to make the project measurable

The strongest demo is the review-and-approval workflow, not merely summarization.

P14 — DocuFlow — Smart Form Filling Assistant

Domain	Automation + AI	Difficulty	6/10
Estimated potential	87/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Agentic AI: Autonomous Enterprise Operations.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

People repeatedly enter the same profile data into forms. Manual copying wastes time and increases typing errors, especially when forms have different layouts.

Use case / proposed solution

Maintain a user profile, detect fields in an uploaded form and propose mappings. The user confirms sensitive or ambiguous fields before any submission.

Why do we need it? The project demonstrates useful browser/document automation while retaining a human approval boundary.	Unique approach Parse form labels, normalize field names and map them to structured profile fields. Use confidence thresholds and confirmation for sensitive values.
---	--

Prototype workflow

- Create profile
- Upload/open form
- Detect fields
- Map fields
- Show confidence
- User confirms
- Fill/export

Minimum viable prototype (MVP)

- Profile store
- PDF form parser
- Field mapping
- Confidence UI
- Manual correction

Advanced extensions

- Browser automation sandbox
- Reusable mappings
- Multiple profiles
- Document templates

- Accessibility features

Suggested stack

React; FastAPI; PDF/OCR parser; controlled browser automation for a sandbox; SQLite.

Live demo plan

Use two differently designed forms asking for the same information. Show how the mapping adapts.

Risks / limitations

Never submit real forms automatically in the prototype. Keep external actions behind explicit confirmation.

How to make the project measurable

A visible approval checkpoint makes the automation trustworthy and easier to evaluate.

P15 — SafeRoute — Multi-Factor Route Planner

Domain	AI + GIS + Automation	Difficulty	6/10
Estimated potential	88/100	Source type	Adapted project idea inspired by the listed geospatial challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — SpaceTech & Satellite Applications: Geospatial Intelligence for Smart Decision-Making; adapted to route planning.

Adapted project idea inspired by the listed geospatial challenge theme; not an official challenge title.

What is the problem?

Shortest route is not always the route a user wants. Users may prioritize distance, travel time, accessibility, road type, congestion or other available factors.

Use case / proposed solution

A map interface where users adjust route priorities and see alternative routes with an explanation of the tradeoffs.

Why do we need it? The project demonstrates multi-objective decision support rather than a basic map. It also makes the weighting logic visible to the user.	Unique approach Generate candidate routes, normalize criteria and compute a weighted score. Show the contribution of each factor and allow users to change weights.
--	---

Prototype workflow

- Select origin/destination
- Choose priorities
- Generate candidate routes
- Score routes
- Display tradeoffs
- Allow weight changes
- Compare outcomes

Minimum viable prototype (MVP)

- Map
- Two/three route candidates
- Weight controls
- Route score breakdown
- Comparison panel

Advanced extensions

- Accessibility profiles
- Weather layer
- Time-dependent routing

- Crowd feedback
- Offline map cache

Suggested stack

React; MapLibre; OpenStreetMap-based routing; FastAPI; PostGIS optional.

Live demo plan

Change the user preference from fastest to accessibility-focused and show how the route ranking changes.

Risks / limitations

External routing data may be incomplete. Clearly label the data source and avoid safety guarantees.

How to make the project measurable

The interactive weight slider provides a strong live demonstration of explainable decision support.

P16 — OpsPilot — AI Workflow Automation Builder

Domain	Automation + Agentic AI	Difficulty	6/10
Estimated potential	92/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Agentic AI: Autonomous Enterprise Operations; Responsible & Trustworthy Agentic AI.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

Workflow automation tools often require users to understand technical configuration. Small teams may know the desired process but not how to express it as connected automation steps.

Use case / proposed solution

User describes a workflow in natural language. The system converts it into a structured plan of triggers, actions, conditions and approval points. Execution happens only after review.

Why do we need it?

This combines natural-language interaction with real workflow structure and a human control layer.

Unique approach

Use an LLM to produce a constrained workflow JSON schema. Validate the schema, render it visually, require approval, then execute only safe demo actions.

Prototype workflow

- Describe workflow
- Generate structured plan
- Validate schema
- Render graph
- User edits/approves
- Run sandbox actions
- Show execution log

Minimum viable prototype (MVP)

- Natural-language input
- Workflow graph
- 3–5 action types
- Validation
- Approval gate
- Execution log

Advanced extensions

- Multi-agent planning
- Connector marketplace
- Retry policies

- Secrets vault
- Versioning
- Simulation mode

Suggested stack React; FastAPI; workflow engine; JSON schema validation; LLM; SQLite/PostgreSQL.	Live demo plan Ask for a workflow such as “when a new support ticket arrives, classify it, assign a team and notify the team.” Show generated steps, edit one, approve and run a sandbox execution.
---	---

Risks / limitations

Never allow arbitrary code execution. Use an allowlisted action set and simulated connectors for the hackathon.

How to make the project measurable

The project has strong demo potential because judges can see natural language become a visible, validated workflow.

P17 — AlertHub — Smart Notification Prioritizer

Domain	AI + Automation	Difficulty	5/10
Estimated potential	83/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Human–AI Collaboration: Adaptive AI for Personalized Collaboration.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Users can receive dozens of notifications from email, chat, monitoring tools and applications. Important alerts can be buried by duplicates or low-value messages.

Use case / proposed solution

Aggregate synthetic notifications, group duplicates, score urgency/actionability and present a focused inbox. Users can change priorities and see why an item moved.

Why do we need it?

Notification overload is a practical human-AI collaboration problem. The system should reduce noise without hiding information.

Unique approach

Cluster similar messages, apply user-specific rules and use optional semantic similarity for grouping. Preserve original messages and provide “why prioritized” evidence.

Prototype workflow

- Ingest notifications
- Normalize metadata
- Group duplicates
- Score urgency/actionability
- Personalize ranking
- Show rationale
- Collect feedback

Minimum viable prototype (MVP)

- Synthetic notification generator
- Grouping
- Priority ranking
- Filters
- Rationale panel

Advanced extensions

- Calendar context
- Learning from feedback
- Cross-app dashboard
- Quiet hours

- Team alert mode

Suggested stack

React; FastAPI; Python; embeddings/rules; SQLite.

Live demo plan

Inject 30 notifications with duplicates and mixed urgency. Show how the focused inbox reduces clutter while retaining access to the originals.

Risks / limitations

A low-priority alert should remain retrievable. Avoid destructive filtering.

How to make the project measurable

Personalization plus transparent ranking can make a simple notification project feel substantially more intelligent.

P18 — AccessGuard — Risk-Based Login Simulator

Domain	Cybersecurity + FinTech	Difficulty	6/10
Estimated potential	90/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Cyber Security: Privacy-Preserving Digital Identity; Building Trust in the Digital Future.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

Authentication can be too weak when context is ignored and too inconvenient when every login receives the same friction. A risk-based system can adjust the challenge according to context.

Use case / proposed solution

Simulate login attempts using synthetic signals such as new device, unusual time, new location and failed attempts. Produce an explainable risk level and select allow, MFA or review.

Why do we need it? The project demonstrates adaptive security while keeping the decision boundary visible and testable.	Unique approach Create a synthetic risk engine with weighted features and optional anomaly detection. Do not use real identity data; focus on the decision flow.
---	--

Prototype workflow

- Generate login attempt
- Extract context
- Calculate risk
- Explain factors
- Choose challenge
- Record decision
- Review history

Minimum viable prototype (MVP)

- Synthetic login generator
- Risk score
- MFA simulation
- Decision explanation
- History

Advanced extensions

- Device fingerprint simulation
- Behavioral baseline
- Privacy-preserving features
- Admin policy editor

Suggested stack

React; FastAPI; Python rules/ML; SQLite.

Live demo plan

Run normal login, new-device login and repeated-failure login. Show how the decision changes and why.

Risks / limitations

Use synthetic identities. Avoid collecting real passwords or biometric data.

How to make the project measurable

A clear decision matrix, interactive scenarios and measurable false-positive/false-negative tradeoffs improve the technical story.

P19 — ScholarshipMatch — AI Eligibility Assistant

Domain	EdTech + FinTech + AI	Difficulty	5/10
Estimated potential	86/100	Source type	Adapted project idea inspired by the listed challenge theme; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — FinTech & Digital Economy: Intelligent Financial Inclusion; adapted to scholarships.

Adapted project idea inspired by the listed challenge theme; not an official challenge title.

What is the problem?

Scholarship requirements can be spread across long documents, and students may not know which criteria they satisfy or which information still needs verification.

Use case / proposed solution

Upload scholarship documents and enter student details. The system maps each eligibility criterion to evidence and labels each as likely eligible, not met, or needs verification.

Why do we need it?

The goal is not to make a final legal/administrative decision but to reduce search effort and make requirements understandable.

Unique approach

Extract eligibility rules into a structured schema, normalize user inputs and evaluate each criterion separately. Show the source evidence for every result.

Prototype workflow

- Upload scholarship document
- Extract criteria
- Enter student profile
- Evaluate criteria
- Show evidence
- Flag uncertain items
- Export checklist

Minimum viable prototype (MVP)

- PDF input
- Criteria extraction
- Profile form
- Criterion-by-criterion checklist
- Evidence view

Advanced extensions

- Deadline tracker
- Document checklist
- Multilingual support
- Institution dashboard

- Change detection

Suggested stack

React; FastAPI; PDF extraction; rules engine; optional LLM.

Live demo plan

Use two fictional scholarships with different criteria and show how the checklist changes.

Risks / limitations

Requirements change and documents may be ambiguous. Always show source text and mark uncertain cases for verification.

How to make the project measurable

Evidence-linked eligibility checks are easier to trust than an opaque “you qualify” answer.

P20 — CrisisAssist — Emergency Response Dashboard

Domain	Agentic AI + Automation	Difficulty	6/10
Estimated potential	92/100	Source type	Adapted project idea inspired by the listed challenge themes; not an official challenge title.

Source / challenge direction

Hack Celestia 2026 — Agentic AI: Intelligent Crisis Response Agents; Human–AI Collaboration: Human–AI Collaboration for Crisis & Disaster Response.

Adapted project idea inspired by the listed challenge themes; not an official challenge title.

What is the problem?

During a simulated emergency, responders may receive alerts from multiple sources and need a concise picture of what is happening, what is verified and what actions require attention.

Use case / proposed solution

Combine simulated alerts, map points and resource status. Generate an incident summary, priority list and action checklist while clearly separating verified inputs from AI-generated recommendations.

Why do we need it?

The project showcases agentic coordination without allowing the AI to silently perform high-impact actions. Human approval remains central.

Unique approach

Use a small set of agents or modules: ingestion, situation summarization, prioritization and checklist generation. Require source references and approval before any simulated external action.

Prototype workflow

- Ingest simulated alerts
- Normalize events
- Map incidents
- Summarize situation
- Prioritize actions
- Human reviews
- Execute sandbox action
- Record audit log

Minimum viable prototype (MVP)

- Map/dashboard
- Synthetic alert feed
- Incident grouping
- Action checklist
- Source/evidence panel
- Approval gate

Advanced extensions

- Resource allocation simulation
- Multi-agent coordination
- Time-series replay
- What-if scenarios
- After-action report

Suggested stack

React; FastAPI; MapLibre; PostGIS optional; rules/LLM; SQLite/PostgreSQL.

Live demo plan

Replay a fictional flood scenario with multiple alerts arriving over time. Show how the dashboard updates the situation and produces an approval-ready action list.

Risks / limitations

Keep the entire system simulated. Do not present AI recommendations as emergency instructions for real incidents.

How to make the project measurable

A replayable scenario, source-aware recommendations and human approval provide a strong end-to-end demonstration.

23. Final Summary & Comparison

Use this table for quick team discussion. The source column intentionally uses the challenge direction rather than presenting the proposed title as an official problem statement.

ID	Project	Domain	Diff.	Potential*	Source direction
P01	Explainable Phishing Detection	Cybersecurity + AI	5/10	90/100	Cyber Security: Autonomous Threat Detection & Response; Building Trust in the Digital Future.
P02	AI Security Test Lab	Cybersecurity + AI	6/10	93/100	Cyber Security: Securing the AI-Driven Digital Ecosystem.
P03	AI Security Log Analyzer	Cybersecurity + Automation	6/10	89/100	Cyber Security: Autonomous Threat Detection & Response.
P04	Scam Message Analyzer	Cybersecurity + GenAI	5/10	88/100	Cyber Security: Building Trust in the Digital Future; Generative AI: Responsible & Trustworthy Generative AI.
P05	AI Spending Coach	FinTech + AI	5/10	88/100	FinTech & Digital Economy: Intelligent Financial Decision Support.
P06	Transaction Anomaly Detector	FinTech + AI	6/10	91/100	FinTech & Digital Economy: AI-Driven Financial Risk & Fraud Intelligence.
P07	Smart Invoice Extractor	FinTech + Automation	5/10	86/100	FinTech & Digital Economy: Intelligent Financial Decision Support; adapted toward document automation.
P08	Adaptive AI Study Planner	EdTech + GenAI	5/10	87/100	Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to education.
P09	College Knowledge Assistant	EdTech + GenAI	5/10	89/100	Generative AI: AI-Powered Digital Knowledge Systems; adapted to education.
P10	Resume-to-Job Analyzer	EdTech + AI	5/10	85/100	Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to careers.
P11	Weak Topic Detector	EdTech + AI	6/10	88/100	Human–AI Collaboration: Adaptive AI for Personalized Collaboration; adapted to learning analytics.
P12	Complaint & Ticket Automation	Automation + AI	5/10	86/100	Agentic AI: Autonomous Enterprise Operations.
P13	Meeting-to-Task Automation	Automation + GenAI	5/10	84/100	Agentic AI: Autonomous Enterprise Operations; Human–AI Collaboration: AI as a Collaborative Intelligence Partner.
P14	Smart Form Filling Assistant	Automation + AI	6/10	87/100	Agentic AI: Autonomous Enterprise Operations.
P15	Multi-Factor Route Planner	AI + GIS + Automation	6/10	88/100	SpaceTech & Satellite Applications: Geospatial Intelligence for Smart Decision-Making; adapted to route planning.
P16	AI Workflow Automation Builder	Automation + Agentic AI	6/10	92/100	Agentic AI: Autonomous Enterprise Operations; Responsible & Trustworthy Agentic AI.
P17	Smart Notification Prioritizer	AI + Automation	5/10	83/100	Human–AI Collaboration: Adaptive AI for Personalized Collaboration.
P18	Risk-Based Login Simulator	Cybersecurity + FinTech	6/10	90/100	Cyber Security: Privacy-Preserving Digital Identity; Building Trust in the Digital Future.
P19	AI Eligibility Assistant	EdTech + FinTech + AI	5/10	86/100	FinTech & Digital Economy: Intelligent Financial Inclusion; adapted to scholarships.

ID	Project	Domain	Diff.	Potential*	Source direction
P20	Emergency Response Dashboard	Agentic AI + Automation	6/10	92/100	Agentic AI: Intelligent Crisis Response Agents; Human–AI Collaboration: Human–AI Collaboration for Crisis & Disaster Response.

*Potential is an unofficial prototype-potential estimate, not an official judging score.

24. Suggested Team Shortlisting Matrix

Before committing to one project, score your own team on build time, access to data, demo reliability, technical depth and ability to measure results. This is a planning matrix, not an external ranking.

Criterion	Question to ask	Evidence to prepare
Buildability	Can we finish a working MVP during the event?	Feature cut-list + milestone plan
Data	Can we legally and safely obtain demo data?	Synthetic/public dataset + sample size
Technical depth	Is there more than a UI or API wrapper?	Architecture + algorithm/model + evaluation
Demo clarity	Can a judge understand the value in 2–3 minutes?	Before/after or scenario replay
Measurability	Can we report a meaningful metric?	Precision/recall, latency, task success, etc.
Trust & safety	Does the system expose uncertainty and human control?	Evidence, confidence, audit log, approval gate

Final note on sources

Hack Celestia’s published challenge page is the primary source for the challenge-theme names used throughout this handbook. SIH 2026 is referenced as a separate benchmark for the broader software-hackathon context. If you decide to submit to SIH, replace any adapted idea with the exact official SIH problem-statement title/code after verifying it in the current catalogue. ■cite■turn1search1■turn1search0■